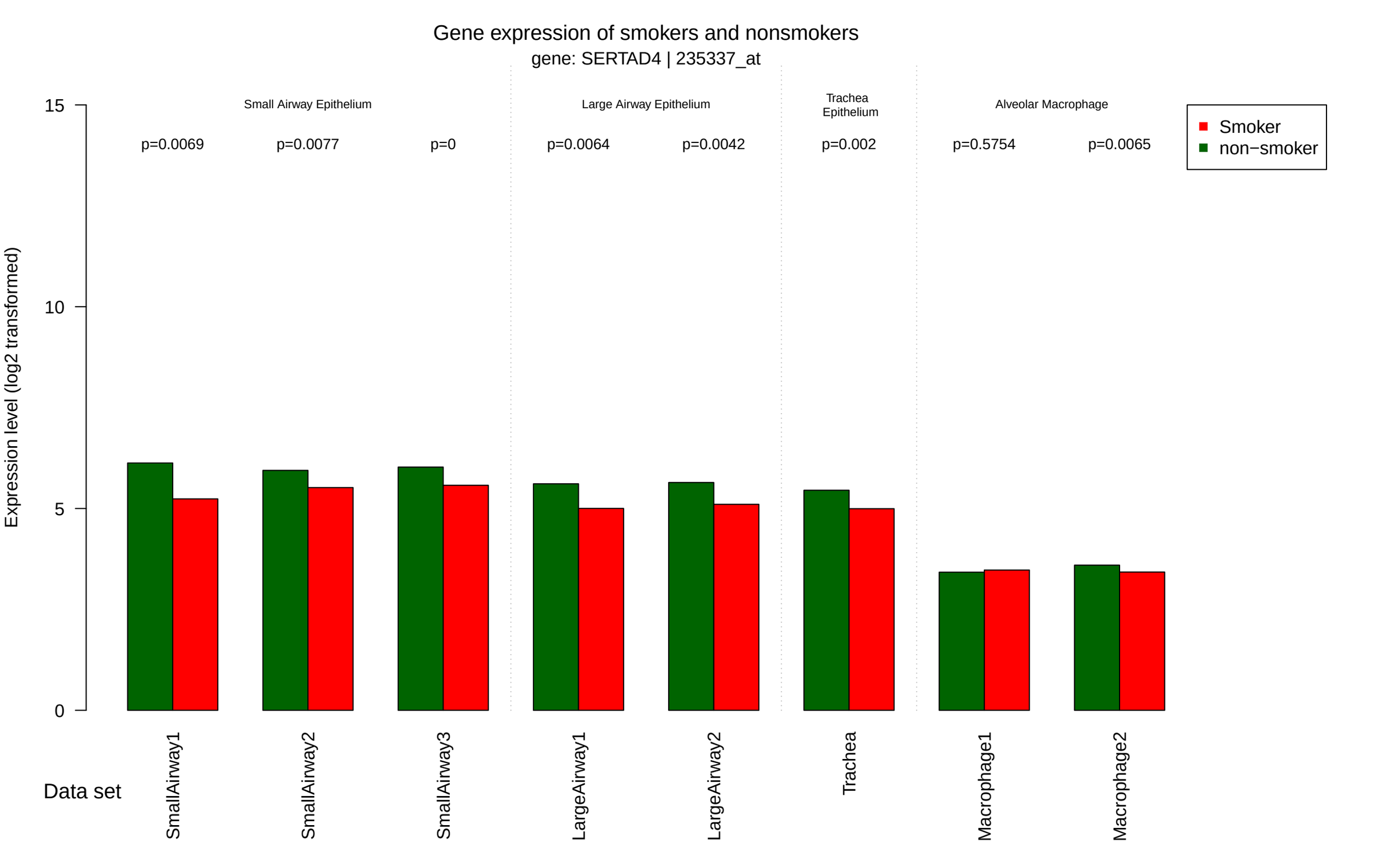
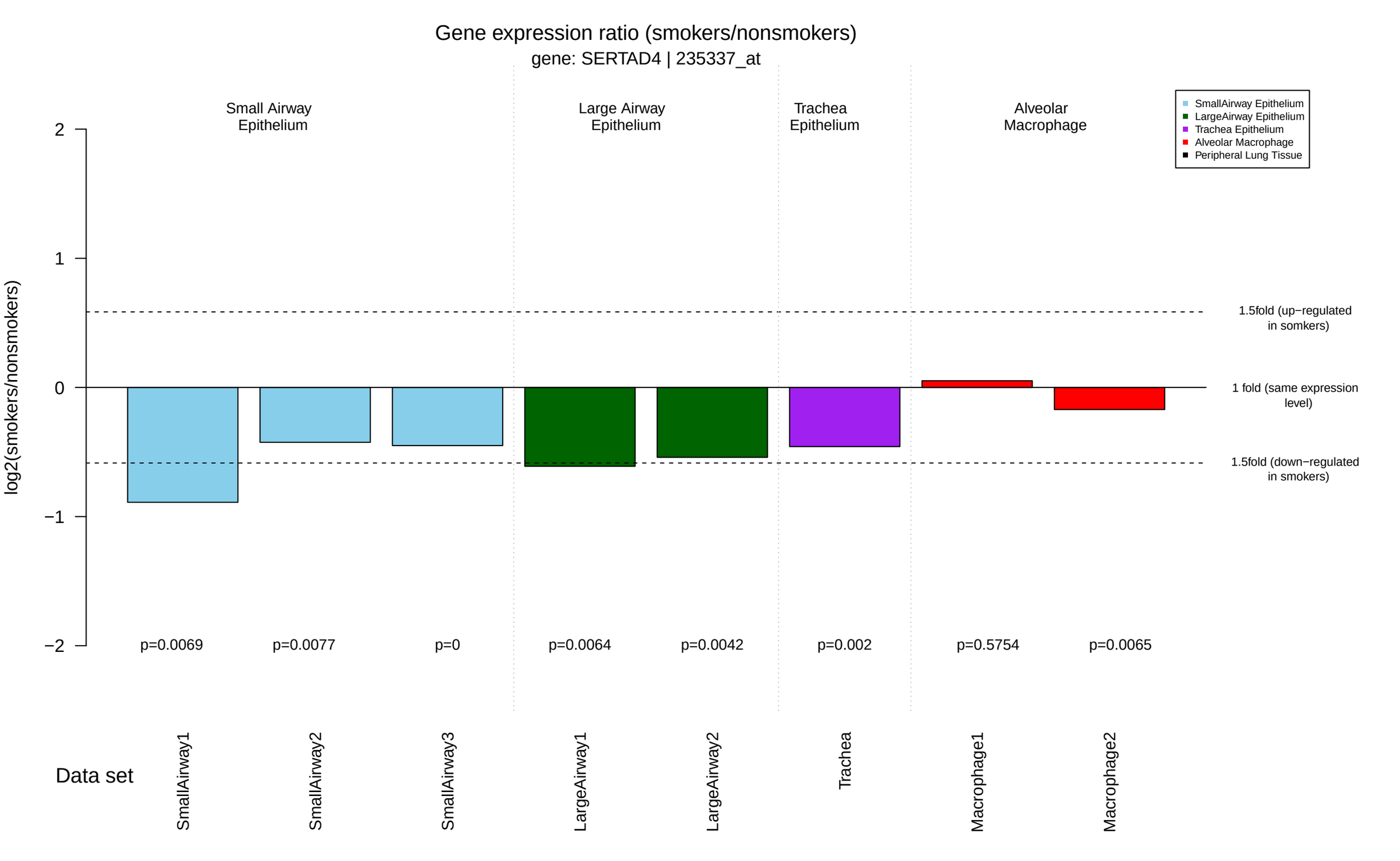
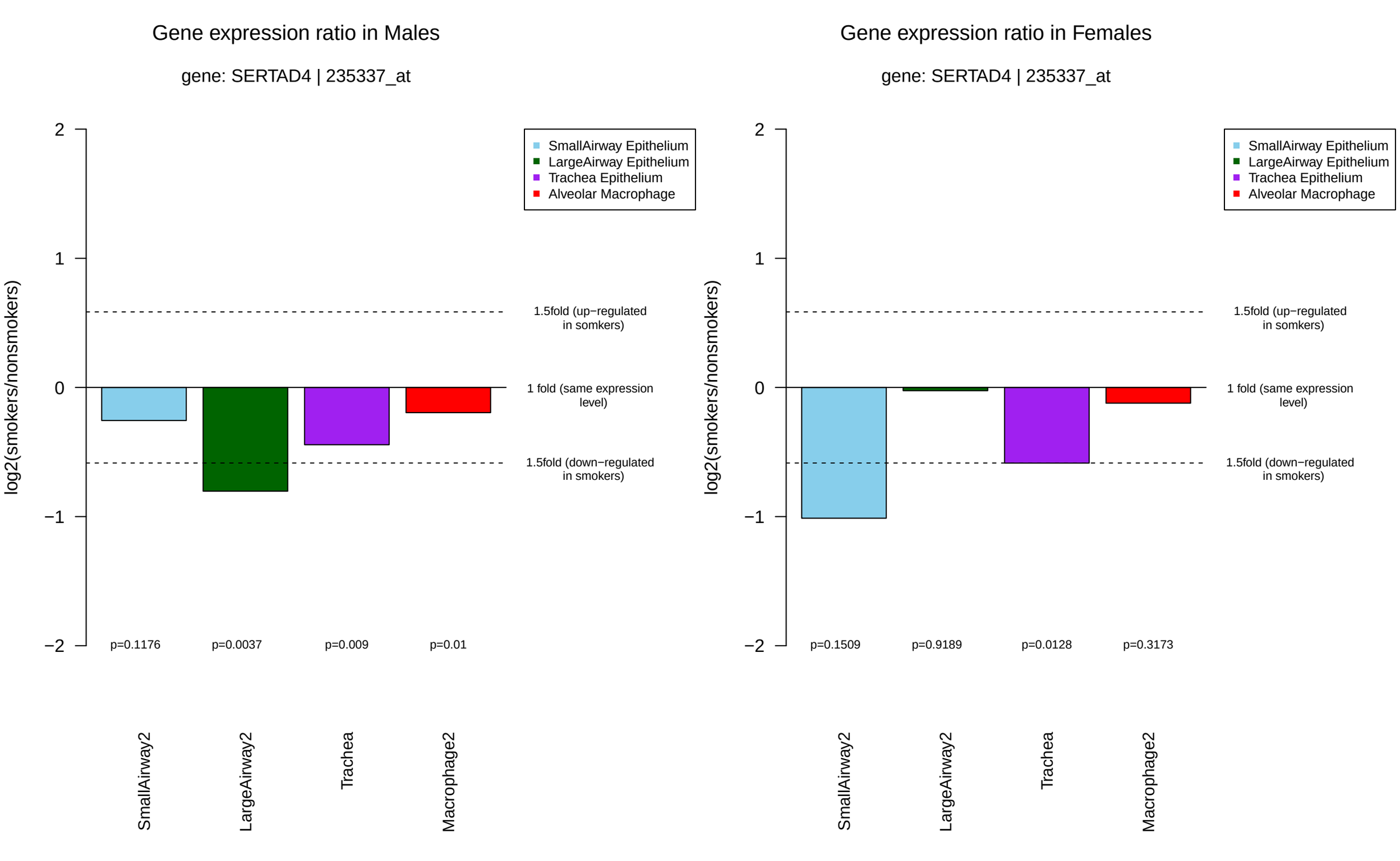
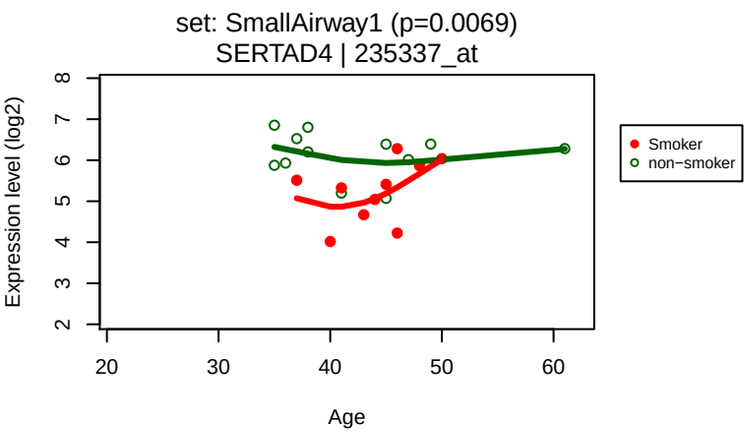
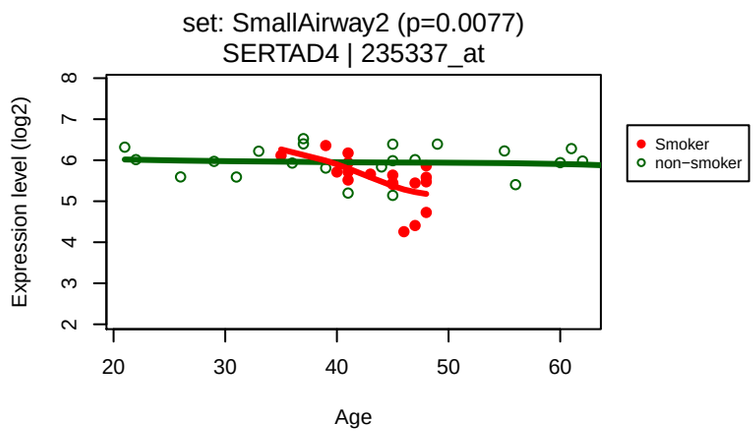
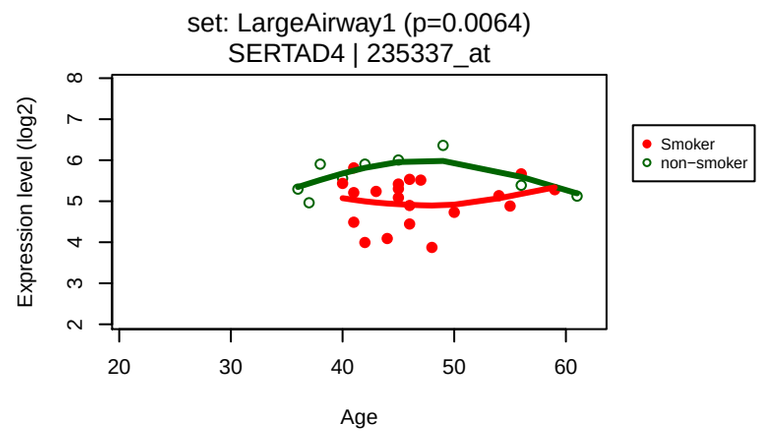

Gene expression of smokers and nonsmokers
gene: SERTAD4 | 235337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0069
p=0.0077
p=0
p=0.0064
p=0.0042
p=0.002
p=0.5754
p=0.0065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

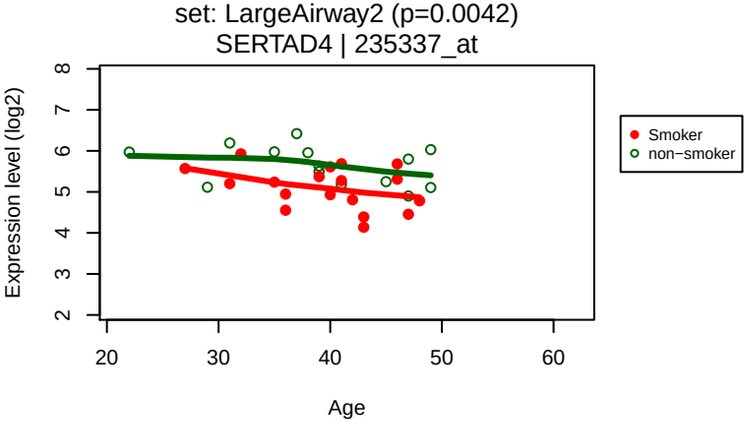

Gene expression ratio (smokers/nonsmokers)
gene: SERTAD4 | 235337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0069
p=0.0077
p=0
p=0.0064
p=0.0042
p=0.002
p=0.5754
p=0.0065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

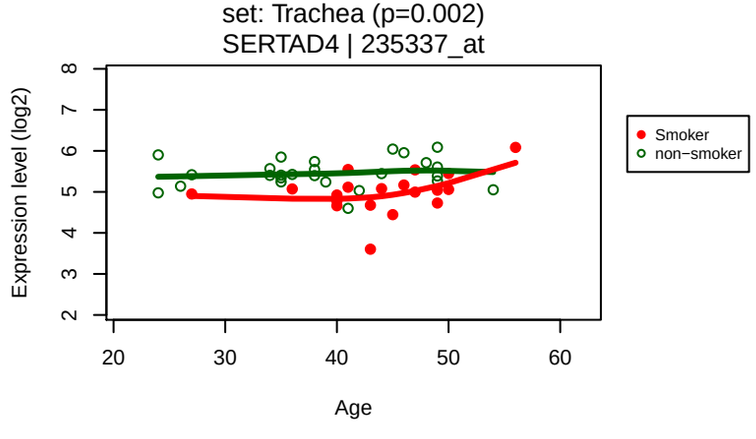

Gene expression ratio in Males
gene: SERTAD4 | 235337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1176
p=0.0037
p=0.009
p=0.01
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERTAD4 | 235337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1509
p=0.9189
p=0.0128
p=0.3173
SmallAirway2
LargeAirway2
Trachea
Macrophage2

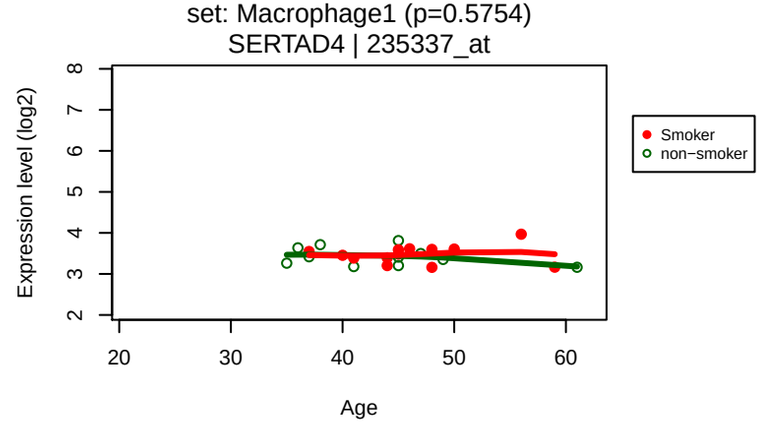

set: SmallAirway1 (p=0.0069)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker

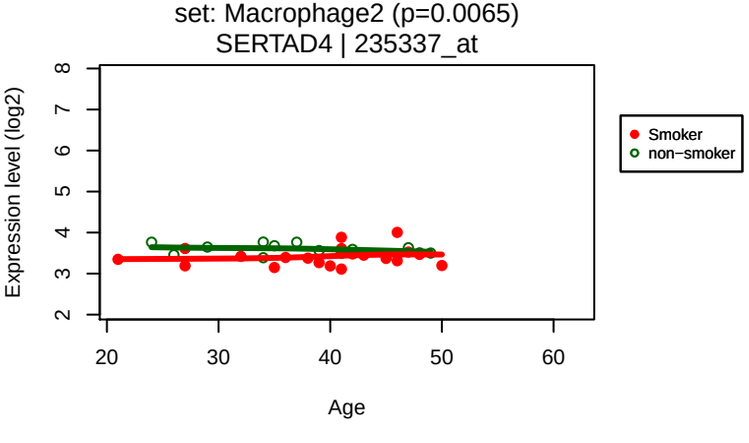

set: SmallAirway2 (p=0.0077)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker

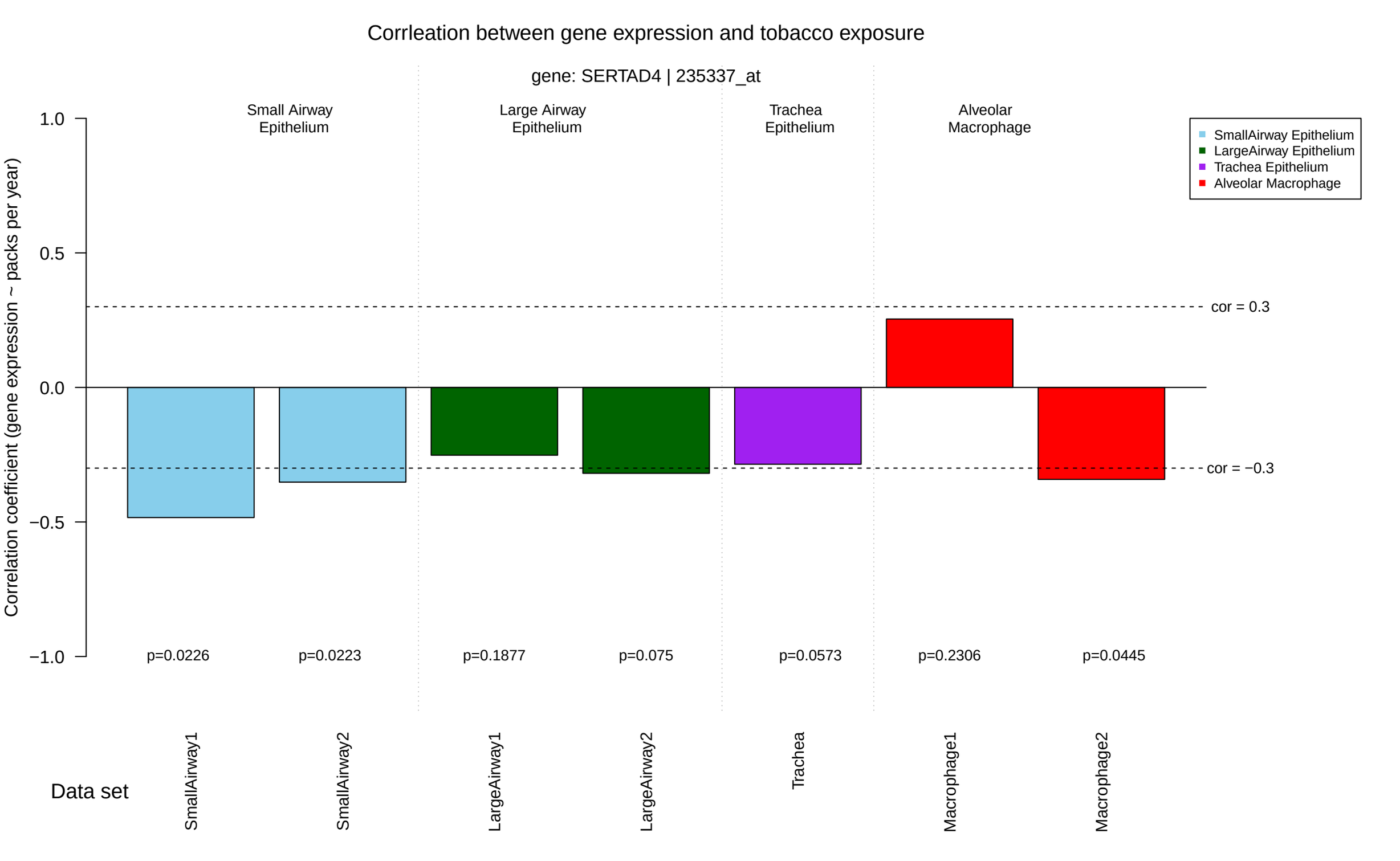

set: LargeAirway1 (p=0.0064)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0042)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.002)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5754)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0065)
SERTAD4 | 235337_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERTAD4 | 235337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0226
p=0.0223
p=0.1877
p=0.075
p=0.0573
p=0.2306
p=0.0445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2